

Toward Closed-Form American Option Prices: Phase-Affine Boundaries and the Occupation Resolvent

Quadrature-free American pricing under BSM with dividend yield, from the one-driver
complex GBM program

Complex-Itô Random-Walk Research Project

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Phase 15 priced American calls and puts under Black-Scholes-Merton with dividend yield q as European Black-Scholes plus one early-exercise- premium (EEP) integral over the collocated logarithmic-spiral boundary — leaving one numerical quadrature in the formula. Phase 16 removes it. We prove and validate (to machine precision, worst error 4.9×10^{-14} on randomized trials) a closed form for the discounted occupation integral of drifted Brownian motion, $J(\tau; a, m, \lambda) = \int_0^\tau e^{-\lambda s} \Phi((a+ms)/\sqrt{s}) ds$, via the drift-diffusion resolvent and its finite-horizon tail. Consequently every phase-affine boundary ($\ln b$ affine in time to maturity) admits a fully closed-form EEP, and **multipiece phase-affine (PPA) boundaries** — sequential value-matching collocation of the knot values, perpetual pinning at long maturity — give American prices as finite sums of closed-form terms: no numerical integration anywhere. On a reduced grid (18 configurations, $T \leq 3$) the recommended $m = 10$ configuration attains RMSE 0.0079 against the reference boundary solver, versus 0.164 for Barone-Adesi-Whaley, at ~ 0.07 s per option. Two structural findings accompany the method: the boundary’s approach to the perpetual limit is generically *not* a pure exponential (effective decay rates drift with the observation window, most extremely at $r = q$, where the linearized boundary-memory kernel $\propto (r - q)$ degenerates), which justifies perpetual pinning over extrapolated decay; and a “Tier B” attempt to replace even the collocation solves by explicit parameter surfaces currently degrades accuracy (RMSE ≈ 0.05) and is documented as open. We state precisely what is and is not claimed: no closed form for the true free boundary is asserted.

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1 Introduction

Phase 15 (Complex–Itô Random-Walk Research Project 2026a) established the exact early-exercise-premium machinery for American options under BSM with dividend yield q (Kim 1990; Jacka 1991; Carr et al. 1992), a stable causal reference solver for the nonlinear Volterra boundary equation, and the collocated logarithmic-spiral pricing formula: European Black–Scholes plus one one-dimensional integral over an explicit boundary. The remaining quadrature is the only numerical object left in the formula. Phase 16 removes it.

The key is classical in optimal-stopping theory but appears not to have been deployed as a *pricing primitive* in this form: the EEP integrand is a Gaussian probability,

$$\Phi(d_1(S, b(\tau - s), s)) = \mathbb{P}(W_s + ms \geq -a), \quad (1)$$

when the boundary is **phase-affine** — $\ln b(u) = A - \gamma u$ in time to maturity u — because then d_1 is affine in s under the $\sqrt{s}$ scaling. The discounted time-integral of such probabilities is the **occupation resolvent** of drifted Brownian motion, and it has a closed form (Section 2). Multipiece phase-affine (PPA) boundaries — piecewise-straight phase barriers in the Phase 4 complex-GBM picture (Complex–Itô Random-Walk Research Project 2026b) — inherit the closed form piece by piece (Section 3).

The knot values of the PPA boundary are fixed by sequential value-matching collocation (one scalar solve per knot, causal from maturity), with the last knot **pinned at the perpetual boundary** for long maturities. The pin is not cosmetic: Section 5 documents that the true boundary’s approach to the perpetual limit is generically not exponential — effective decay rates fitted on successively later windows *drift* (e.g. $0.142 \rightarrow 0.120 \rightarrow 0.101$ for the $r = q = 0.05$ put), and at $r = q$ the linearized boundary-memory kernel is proportional to $r - q$ and degenerates entirely. Extrapolating a decay law into the far maturity is therefore exactly the wrong closure; pinning at the known limit is the right one.

Headline. On the reduced grid (18 configurations $\times$ 3 spots, $T \in \{0.5, 1, 3\}$) the $m = 10$ PPA formula attains RMSE 0.0079 / MAE 0.054 against the reference, versus 0.164/0.437 for Barone-Adesi–Whaley — with **zero numerical integration**: every price is European Black–Scholes plus a finite sum of Φ -evaluations (Table 1). The MAE is dominated by the $T = 3$ configurations, where the reference solver itself carries an uncertainty of the same order (Section 7).

1.0.1 Novelty boundary

Claimed as new: the closed-form occupation integral J as an EEP building block and its machine-precision validation; the multipiece phase-affine boundary family with sequential collocation and perpetual pinning as a quadrature-free American pricing formula; the documented non-exponential perpetual approach and the $r = q$ degeneracy of the linearized kernel; the honest Tier B negative result. *Not claimed:* the EEP representation itself, the anchors, the perpetual closed form, multipiece exponential boundaries as an idea (Ju’s construction uses them with local quadratic pricing (Ju 1998); our per-piece prices are exact given the piecewise boundary), or any closed form for the true free boundary — the constant-coefficient problem remains open.

2 The closed-form occupation integral

2.1 Statement

For $\lambda > 0$, $a, m \in \mathbb{R}$, define

$$J(\tau; a, m, \lambda) = \int_0^\tau e^{-\lambda s} \Phi\left(\frac{a + ms}{\sqrt{s}}\right) ds. \quad (2)$$

Writing $X_s = W_s + ms$ and $\nu = \sqrt{m^2 + 2\lambda}$, the perpetual occupation resolvent of X against the half-line $\{X \geq -a\}$ is

$$R(a; \lambda, m) = \int_0^\infty e^{-\lambda s} \mathbb{P}(X_s \geq -a) ds = \begin{cases} \frac{1}{\lambda} - \frac{e^{-(m+\nu)a}}{\nu(\nu+m)}, & a \geq 0, \\ \frac{e^{(\nu-m)a}}{\nu(\nu-m)}, & a < 0, \end{cases} \quad (3)$$

and the finite-horizon integral is the perpetual resolvent minus the Markov tail, which evaluates through two Gaussian convolutions of piecewise-exponential functions:

$$\boxed{J(\tau; a, m, \lambda) = R(a; \lambda, m) - e^{-\lambda\tau} \left[\frac{1}{\nu(\nu+m)} + \frac{1}{\nu(\nu-m)} \right] \Phi\left(\frac{a+m\tau}{\sqrt{\tau}}\right) + \frac{e^{-(m+\nu)a}}{\nu(\nu+m)} \Phi\left(\frac{a-\nu\tau}{\sqrt{\tau}}\right) - \frac{e^{(\nu-m)a}}{\nu(\nu-m)} \Phi\left(\frac{-a-\nu\tau}{\sqrt{\tau}}\right)}. \quad (4)}$$

Checks: $J(0^+) = 0$ (the three Φ -terms reconstruct R), $J(\infty) = R$, $J(\cdot; a \rightarrow +\infty) = (1 - e^{-\lambda\tau})/\lambda$, $J(\cdot; a \rightarrow -\infty) = 0$. Products of exponentials with extreme arguments and tiny Φ -values are evaluated through $\log \Phi$ to avoid overflow/cancellation.

2.2 Validation

Against adaptive quadrature (tolerances 10^{-13}) of the defining integral, 60 randomized trials ($a \in [-4, 4]$, $m \in [-2, 2]$, $\lambda \in [0.01, 0.3]$, $\tau \in [0.01, 4]$) agree to worst absolute error 4.9×10^{-14} ; all four analytic limits hold to $\leq 4 \times 10^{-8}$ (the $\tau \rightarrow \infty$ check at $\tau = 400$). Fixed-order Gauss–Legendre quadrature of the defining integral converges only slowly on thin transition layers (errors $\sim 10^{-6}$ at 160 nodes in corners), which makes the closed form not merely fast but *more accurate than naive quadrature*.

3 Phase-affine boundaries and the PPA formula

3.1 Single piece

Let the boundary be phase-affine in time to maturity, $\ln b(u) = A - \gamma u$. Then in ?@eq-eep of Phase 15 the quantities d_1, d_2 at elapsed time s become

$$d_i = \frac{x + \mu_i s}{\sigma \sqrt{s}}, \quad x = \ln S - A + \gamma \tau, \quad \mu_1 = r - q + \frac{1}{2}\sigma^2 - \gamma, \quad \mu_2 = r - q - \frac{1}{2}\sigma^2 - \gamma, \quad (5)$$

so the whole premium is two occupation integrals:

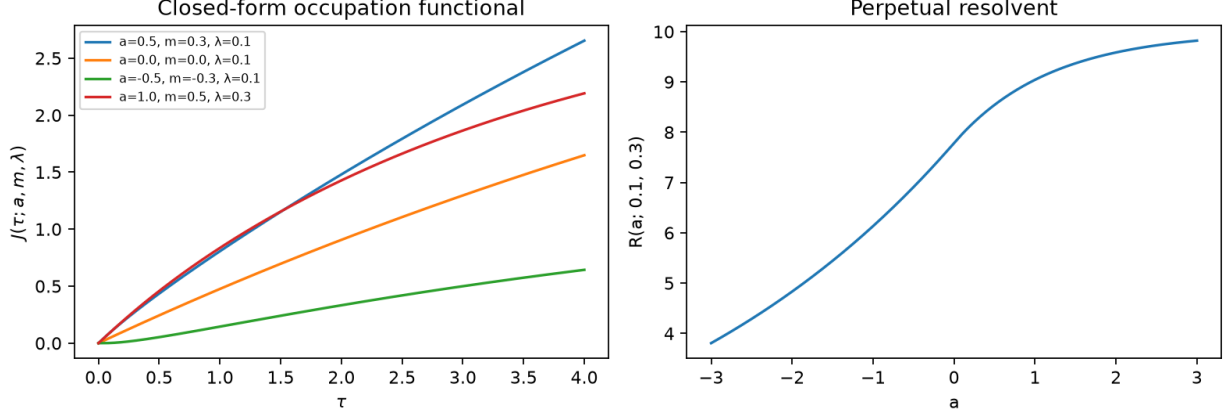


Figure 1: Left: $J(\tau; a, m, \lambda)$ for four parameter triples. Right: the perpetual resolvent $R(a; 0.1, 0.3)$ across the $a = 0$ branch point.

$$\text{EEP}^{\text{call}} = qS J(\tau; x/\sigma, \mu_1/\sigma, q) - rK J(\tau; x/\sigma, \mu_2/\sigma, r), \quad (6)$$

with the put obtained from $\Phi(-d) = 1 - \Phi(d)$:

$$\text{EEP}^{\text{put}} = rK \left[\frac{1-e^{-r\tau}}{r} - J(\tau; x/\sigma, \mu_2/\sigma, r) \right] - qS \left[\frac{1-e^{-q\tau}}{q} - J(\tau; x/\sigma, \mu_1/\sigma, q) \right]. \quad (7)$$

These agree with the Phase 15 quadrature pipeline to $\leq 4.9 \times 10^{-14}$ on randomized piece parameters (Section 1.0.1).

3.2 Multipiece boundaries, collocation, and the price

Let $0 = u_0 < u_1 < \dots < u_m = T$ (1.5-clustered knots) and let the log- boundary be affine on each piece, continuous at the knots: $\ln b(u) = A_j - \gamma_j u$ on $(u_{j-1}, u_j]$, with $v_j = \ln b(u_j)$ and $v_0 = \ln b(0^+)$ fixed at the maturity anchor $b(0^+) = K \max(1, r/q)$ (call) or $K \min(1, r/q)$ (put). Causality of the Volterra equation makes the collocation **sequential**: with $v_1, \dots, v_{j-1}$ known, the piece- j slope $\gamma_j = (v_{j-1} - v_j)/(u_j - u_{j-1})$ is a function of v_j alone, and v_j solves value matching at u_j — one scalar root-find per knot. For $T \geq 1.5$ the last knot is pinned at the perpetual limit, $v_m = \ln b_\infty$ (Section 5).

The price is then European Black-Scholes plus a sum of closed-form piece contributions: each piece contributes the difference $J(s_j^+) - J(s_j^-)$ of Eq. 4 at the elapsed-time slice $(s_j^-, s_j^+) = (\max(0, \tau - u_j), \tau - u_{j-1})$ with the piece's (A_j, γ_j) in Eq. 5:

$$V_A(S, T) = V_E(S, T) + \sum_{j=1}^m \left[qS \Delta J_q^{(j)}(x_j, m_1^{(j)}, q) - rK \Delta J_r^{(j)}(x_j, m_2^{(j)}, r) \right] \quad (8)$$

for calls (puts via Eq. 7), where every $\Delta J^{(j)}$ is an explicit combination of Φ -evaluations. **No numerical integration appears anywhere**: the only numerics are m one-dimensional collocation root-finds, ~ 0.07 s total at $m = 10$.

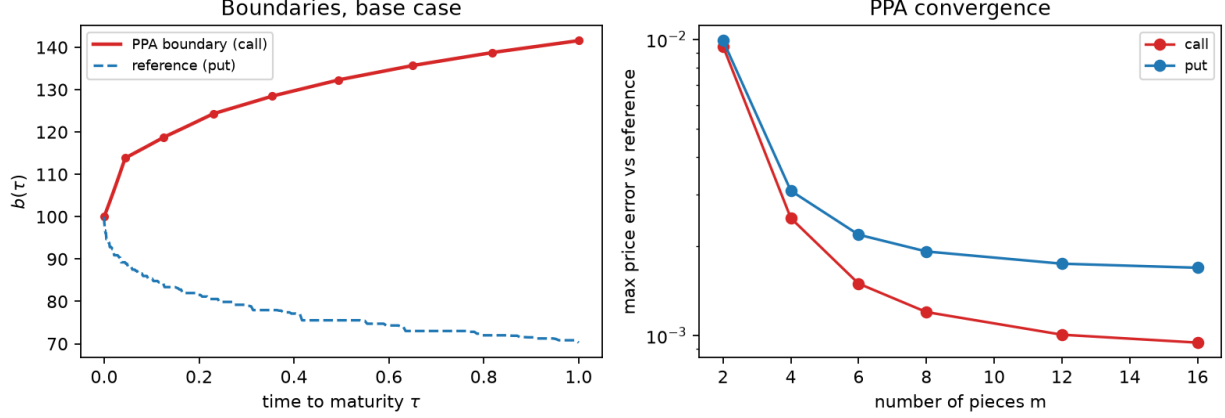


Figure 2: Left: PPA boundary (call, knots marked) against the reference put boundary, base case. Right: error vs m .

4 Numerical results

4.1 Convergence and the reduced grid

Convergence in m on the base case ($r = q = 0.05$, $\sigma = 0.2$, $T = 1$): call errors $0.0098 \rightarrow 0.0018$ and put errors $0.0109 \rightarrow 0.0052$ as m runs $2 \rightarrow 16$ (Figure 2, right). The recommended configuration $m = 10$ with perpetual pinning for $T \geq 1.5$, on the reduced grid (3 rate pairs $\times \sigma \in \{0.2, 0.4\} \times T \in \{0.5, 1, 3\} \times$ both kinds $\times S \in \{90, 100, 110\}$):

Table 1: Reduced-grid errors against the reference solver.

Method	RMSE	MAE
PPA ($m = 10$, pinned)	0.0079	0.054
Phase 15 spiral (collocated)	0.0066	0.039
Barone-Adesi-Whaley	0.164	0.437

The PPA formula matches the Phase 15 spiral in RMSE and removes its quadrature entirely; its MAE is dominated by the $T = 3$ configurations, where the reference boundary itself is uncertain at the same order (Section 7). The extended 90-configuration grid (σ up to 0.6, five spots) gives PPA RMSE 0.016 / MAE 0.167 versus 0.285/1.143 for BAW (Table 2).

4.2 Extended grid

Table 2: Extended 90-configuration benchmark against the reference solver.

Method	RMSE	MAE
PPA ($m = 10$, pinned)	0.0159	0.167
Phase 15 spiral (collocated)	0.0158	0.181
Barone-Adesi-Whaley	0.285	1.143

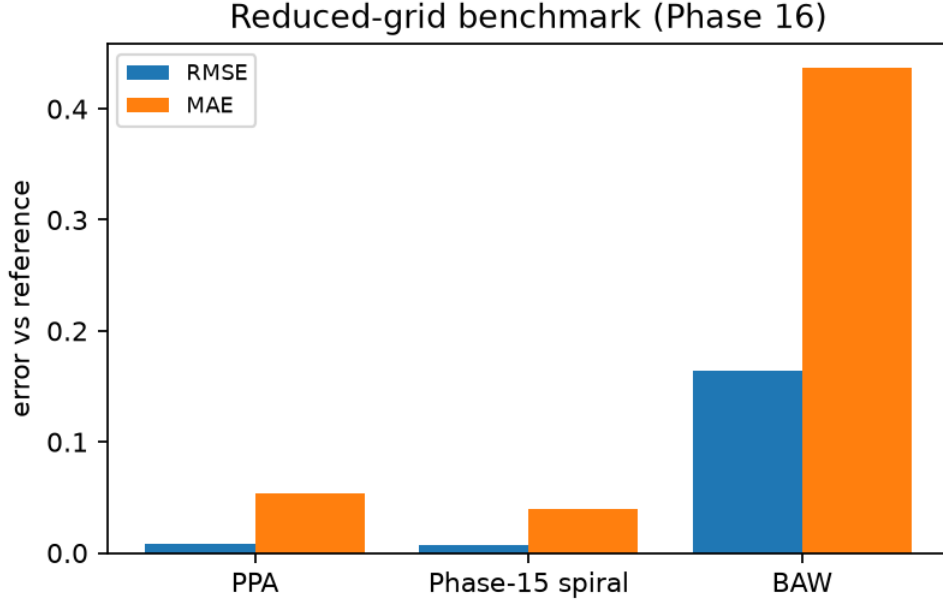


Figure 3: RMSE/MAE against the reference solver on the reduced grid.

5 Structural finding: the perpetual approach is not exponential

Fit $\ln |b(\tau) - b_\infty| = c - \kappa_{\text{eff}}\tau$ on successively later windows of the $T = 16$ reference solutions (Figure 4):

Table 3: Effective decay rates of $|b(\tau) - b_\infty|$ by observation window.

Case	[2, 16]	[6, 16]	[10, 16]
put, $r = q = 0.05$	0.142	0.120	0.101
put, $r = 0.07, q = 0.03$	0.169	0.130	0.169

The rates *drift* with the window: there is no single exponential. The mechanism is visible in the linearization of the value-matching equation around the perpetual solution: with $b(\tau) = b_\infty e^{\varepsilon(\tau)}$ and the exponential ansatz $\varepsilon = Ce^{-\kappa\tau}$, the kernel that carries boundary memory, $\partial f / \partial B$ at (b_∞, b_∞) , is proportional to $(r - q)$ (the Φ -parts cancel by the Black–Scholes delta identity), so at $r = q$ the characteristic equation loses its κ -dependence entirely — the memory enters at higher order and the approach is slower than any rate the first-order theory can name (consistent with the drifting fits). The practical consequence is the design decision of Section 3: **pin the boundary at the known perpetual limit instead of extrapolating a decay law**.

6 Tier B probe: eliminating the collocation solves

A fully evaluation-only formula would replace the m collocation solves by explicit surfaces $\kappa_i(r, q, \sigma, T)$ (Phase 15 spiral parameters) or knot-value surfaces. A 13-term quadratic feature fit $(1, \sigma, r - q, T, \sigma^2, (r - q)^2, T^2, \sigma T, (r - q)T, \sigma(r - q), \sigma^2 T, \sigma T^2, (r - q)\sigma T)$ of the collocated spiral parameters on the reduced grid yields parameter RMS errors 0.08–0.33 (κ_2 worst), and end-to-end the surface-driven price degrades from $\text{RMSE} \approx 0.009$ (collocated) to ≈ 0.05 — an order above the collocated formula, though still $\sim 3\times$ better than BAW. Likely remedies (T -band partitioning, rational bases, direct knot-fraction fits) were probed without crossing the threshold; the direction is documented as open. The operative recipe remains Tier A: m scalar collocations (~ 0.07 s), zero integration.

7 Scope and limitations

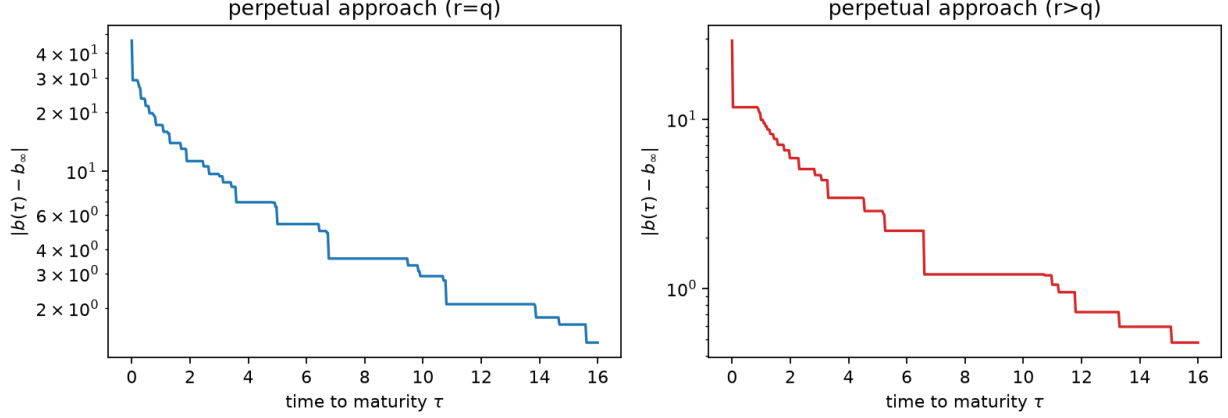


Figure 4: Absolute distance $|b(\tau) - b_\infty|$ for the put at $r = q = 0.05$ (left) and $r = 0.07, q = 0.03$ (right), log scale; the curvature shows the drift of effective rates.

8 Reproducibility

All computations run in the existing **phase3-paper** conda environment (Python 3.12, numpy 2.5.1, scipy 1.18.0, matplotlib 3.11.1); no packages installed. Fixed seed 20260903. Artifacts:

- `phase16_closedform.py` — occupation resolvent, affine/multipiece closed-form EEP, sequential collocation, perpetual pinning, adaptive knot insertion;
- `tests/test_phase16_closedform.py` — 12 tests; full suite 218/218 green;
- `notebooks/build_phase16_closedform.py` → executed `notebooks/phase16_closedform.ipynb` — the computational record behind every number above (figures `phase16_figures/`, tables `phase16_tables/`, extended benchmark CSV);
- `paper/phase16/` — this manuscript, executed notebook copy, tables.

The notebook executes top-to-bottom from a clean kernel in under one minute.

References

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